AIAA SciTech Forum
23-27 January 2023, National Harbor, MD & Online
AIAA SCITECH 2023 Forum
10.2514/6.2023-1794

Conservative Solution Transfer Between Anisotropic Meshes for Adaptive Time-Accurate Hybridized Discontinuous Galerkin Methods

Tomáš Levý *
University of West Bohemia, 301 00 Pilsen, Czech Republic

Georg May †
von Karman Institute for Fluid Dynamics, B-1640 Sint-Genesius-Rode, Belgium

We present a hybridized discontinuous Galerkin (HDG) solver for general time-dependent balance laws. We focus in particular on a coupling of the solution process for unsteady problems with an anisotropic mesh refinement framework. The goal is to properly resolve all relevant unsteady features with the smallest number of mesh elements, and hence to reduce the computational cost of numerical simulations. The crucial step is then to transfer the numerical solution between two meshes since the anisotropic mesh adaptation is producing highly skewed unstructured grids that do not share the same topology as the original mesh where the solution is initially defined. For this purpose, we adopt the Galerkin projection as it preserves the conservation of physically relevant quantities and does not compromise the accuracy of a high-order method. We present numerical experiments verifying these properties of the overall method.

I. Introduction

Over the years, the use of second-order methods such as finite volume schemes has become a standard for the simulation of compressible flows. On the other hand, high-order methods offer superior resolution of the flow field at reduced number of degrees of freedom for a given problem, provided the solution is smooth enough [1]. Since high-speed compressible flow is often accompanied by the presence of shock waves, which in turn disrupts the local regularity of the solution, a finer local resolution is necessary for the high-order method to obtain a reasonably accurate numerical solution. Of course, this need comes with an increased computational cost. Besides the shocks, there could also be other highly anisotropic features of the flow such as boundary layers or wakes in the case of viscous flow simulation. The ideal high-order method then should be adaptive in the sense that it takes advantage of the smooth flow regions where the resolution may not be as fine as in the regions where rapid changes in the solution are present. One such approach is the anisotropic mesh adaptation where the mesh elements align with the features of the flow [2].

The problem at hand is probably even more significant for time-dependent simulations since the shock waves and other anisotropic components of the flow are likely to move over time. In order to fully benefit from the adaptation strategy, the mesh has to be adapted numerous times during the simulation. Such anisotropic mesh refinement generally leads to highly skewed unstructured grids. Hence attention must also be paid to the accuracy of the overall numerical method in order to retain the solution precision in each adaptation step.

An essential pitfall in the remeshing procedure for time-dependent problems is the necessity of interpolating the solution from previous mesh to the adapted mesh that will be used in the subsequent time step. Inappropriate interpolation methods can disrupt conservation of physical quantities of interest, which can ultimately lead to the loss of accuracy. One example is the interpolation method of Grandy [3], which proceeds by mapping from the old mesh to the new one by calculating the volume of intersection of overlapping elements of previous and current meshes. The main drawback of this method is the assumption of constant interpolant across the old-mesh element which results in a highly diffusive approach. Although some conservative interpolation operators coupled with time-dependent anisotropic mesh refinement can be found in the literature [4–6], these are usually designed for locally piecewise linear representation of the solution and the interpolation operator then relies on the Hessian of the solution only [7].

*Graduate Student (levyto@kme.zcu.cz), Department of Mechanics, Technická 8, 301 00 Pilsen, Czech Republic
†Professor (georg.may@vki.ac.be), Aeronautics and Aerospace Department, Waterloosesteenweg 72, B-1640 Sint-Genesius-Rode, Belgium

In the present study, we extend the approach presented in [8] where an anisotropic mesh adaptation methodology is built on hybridized Discontinuous Galerkin (HDG) solver intended to solve boundary-value problems for steady balance laws. The resulting numerical solution on a particular mesh is then represented by a polynomial of degree at most p on each element. For this type of representation, the Galerkin projection, which is optimal in the L^2-norm, proved to be particularly well-suited for interpolation of fields between meshes, [9]. We further adopt the technique of Farrell [10], who applied the Galerkin projection to fields resulting from simulation of incompressible flow [11]. Following this approach, one constructs a geometry, known as the supermesh, defined as the mesh of the intersection polygons of input meshes. On each element of the supermesh, the basis functions of both old and new meshes are, by construction, continuous polynomials, allowing high accuracy evaluation of the projection integrals. Coupling of the time advancement with the mesh adaptation is done by successive generation of meshes after a specified number of time steps whereby the adaptation itself is based on the numerical solution resulting from the previous time step.

The paper is organized as follows. In Sec. II, we briefly review the type of balance laws under consideration. Next, in Sec. III and IV, we discuss the discretization and mesh adaptation methods used in our solver, while Sec. V is dealing with the solution interpolation between meshes. Finally, in Sec. VI, we present some preliminary results based on test cases for scalar convection-diffusion equation and the system of Euler equations showing the conservative and accurate nature of the presented methodology.

II. Governing Equations

In this study, we focus on the solution of time-dependent general (possibly nonlinear) convection-diffusion-reaction systems of equations of the form

$$\partial_t \boldsymbol{w} + \nabla \cdot \left(\boldsymbol{f}_c(\boldsymbol{w}) - \boldsymbol{f}_v(\boldsymbol{w}, \nabla \boldsymbol{w}) \right) = \boldsymbol{s}(\boldsymbol{w}, \nabla \boldsymbol{w}) \tag{1}$$

with appropriate initial and boundary conditions defined on $\partial\Omega$, the boundary of an open bounded domain $\Omega \in \mathbb{R}^d$, $d = 2$. Here, $\boldsymbol{w} \in \mathbb{R}^m$ is the vector of conservative variables, $\boldsymbol{f}_c : \mathbb{R}^m \to \mathbb{R}^{m\times d}$ is the convective flux, $\boldsymbol{f}_v : \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}^{m\times d}$ is the diffusive flux, and $\boldsymbol{s} : \mathbb{R}^m \times \mathbb{R}^{m\times d} \to \mathbb{R}^m$ is the source term, m being the number of conservative variables.

Besides the scalar convection-diffusion equation, this formulation of the balance law also covers the Euler equations with $m = 4$ and

$$\boldsymbol{w} = \begin{pmatrix} \rho \\ \rho \boldsymbol{u} \\ E \end{pmatrix}, \quad \boldsymbol{f}_c = \begin{pmatrix} \rho \boldsymbol{u}^T \\ \rho \boldsymbol{u} \otimes \boldsymbol{u} + P \boldsymbol{I}_d \\ (E + P)\boldsymbol{u}^T \end{pmatrix}, \quad \boldsymbol{f}_v = \boldsymbol{0}, \quad \boldsymbol{s} = \boldsymbol{0} \tag{2}$$

where ρ is the density, $\boldsymbol{u} = (u, v)^T$ is the fluid velocity vector, E is the total energy, and $\boldsymbol{I}_d$ is $d \times d$ identity matrix. Pressure P is related to the conservative variables by the equation of state of an ideal gas in the form

$$P = (\gamma - 1)\left(E - \frac{1}{2}\rho \|\boldsymbol{u}\|_2^2 \right) \tag{3}$$

where γ is the ratio of specific heats and $\gamma = 1.4$ for air.

In case of the presence of shocks in the flow field, the artifical viscosity approach is followed where the diffusive flux is taken to be $\boldsymbol{f}_v = \varepsilon_{\mathrm{AV}} \nabla \boldsymbol{w}$. Here we assume the artificial viscosity coefficient $\varepsilon_{\mathrm{AV}}$ obtained by the shock-capturing method of Hartmann [12] or Moro [13].

III. Discretization

First, let us begin with some notation that will be used throughout the text. Let the computational domain Ω be partitioned into a collection $\mathcal{T}_h$ of nonoverlapping elements such that $\Omega = \bigcup_{\kappa \in \mathcal{T}_h} \kappa$ with $\partial\mathcal{T}_h := \{\partial\kappa \backslash \partial\Omega : \kappa \in \mathcal{T}_h\}$ being the set of all edges of the elements, excluding the boundary. We denote the set of all interior edges of $\mathcal{T}_h$ by $\mathcal{E}_h$, and $\mathcal{E}_h^{\partial}$ contains all boundary edges.

The balance law (1) can be rewritten as a first-order system by introducing an auxiliary variable $\boldsymbol{q}$, which represents the gradient of the solution,

$$\boldsymbol{q} - \nabla \boldsymbol{w} = \boldsymbol{0} \tag{4}$$

$$\partial_t \boldsymbol{w} + \nabla \cdot \left(\boldsymbol{f}_c(\boldsymbol{w}) - \boldsymbol{f}_v(\boldsymbol{w}, \boldsymbol{q}) \right) = \boldsymbol{s}(\boldsymbol{w}, \boldsymbol{q}). \tag{5}$$

In order to obtain a hybridized weak formulation, another additional unknown $\boldsymbol{\lambda}$ is introduced on the element edges. The system of equations is then closed by weakly enforcing continuity of the normal component of convective and diffusive numerical fluxes at the edges. Compared to standard discontinuous Galerkin (DG) methods, this procedure ultimately leads to reduction of the globally coupled degrees of freedom since the method then allows to eliminate the primary variables by a static condensation process and to formulate the global problem in terms of the new variable $\boldsymbol{\lambda}$ only.

Before the weak formulation will be introduced, let us define the discontinuous function spaces in which the solution will be sought. Let $\mathcal{P}^p(D)$ denote a set of polynomials of total degree at most p on some domain D. For a given computational mesh, we will consider the following approximation spaces

$$\Sigma_h := \{\boldsymbol{\tau} \in \left[L^2(\Omega)\right]^{m\times d} : \boldsymbol{\tau}|_\kappa \in [\mathcal{P}^p(\kappa)]^{m\times d}, \forall\kappa \in \mathcal{T}_h\}, \tag{6}$$

$$V_h := \{\boldsymbol{v} \in \left[L^2(\Omega)\right]^{m} : \boldsymbol{v}|_\kappa \in [\mathcal{P}^p(\kappa)]^{m}, \quad \forall\kappa \in \mathcal{T}_h\}, \tag{7}$$

$$\Lambda_h := \{\boldsymbol{\mu} \in \left[L^2(\mathcal{E}_h)\right]^{m} : \boldsymbol{\mu}|_e \in [\mathcal{P}^p(e)]^{m}, \quad \forall e \in \mathcal{E}_h\}. \tag{8}$$

Functions $\boldsymbol{\tau} \in \Sigma_h$, $\boldsymbol{v} \in V_h$, and $\boldsymbol{\mu} \in \Lambda_h$ are then piecewise polynomials of degree p, which can be discontinuous across element faces and vertices, respectively.

Now, the task is to find an approximation $\mathbb{x}_h := (\boldsymbol{q}_h, \boldsymbol{w}_h, \boldsymbol{\lambda}_h) \in \mathbb{X}_h := \Sigma_h \times V_h \times \Lambda_h$ such that

$$\begin{aligned}
0 &= \mathcal{N}(\mathbb{x}_h; \mathbb{y}_h) \\
&:= \left(\boldsymbol{q}_h, \boldsymbol{\tau}_h\right)_{\mathcal{T}_h} + \left(\boldsymbol{w}_h, \nabla\cdot\boldsymbol{\tau}_h\right)_{\mathcal{T}_h} - \langle \boldsymbol{\lambda}_h, \boldsymbol{\tau}_h\cdot\boldsymbol{n}\rangle_{\partial\mathcal{T}_h} \\
&+ \left(\frac{\partial \boldsymbol{w}_h}{\partial t}, \boldsymbol{v}_h\right)_{\mathcal{T}_h} - \left(\boldsymbol{f}_c(\boldsymbol{w}_h) - \boldsymbol{f}_v(\boldsymbol{w}_h, \boldsymbol{q}_h), \nabla \boldsymbol{v}_h\right)_{\mathcal{T}_h} - \left(\boldsymbol{s}(\boldsymbol{w}_h, \boldsymbol{q}_h), \boldsymbol{v}_h\right)_{\mathcal{T}_h} + \left\langle \left(\widehat{\boldsymbol{f}}_c - \widehat{\boldsymbol{f}}_v\right)\cdot\boldsymbol{n}, \boldsymbol{v}_h\right\rangle_{\partial\mathcal{T}_h} \\
&+ \left\langle \left[\!\left[\widehat{\boldsymbol{f}}_c - \widehat{\boldsymbol{f}}_v\right]\!\right], \boldsymbol{\mu}_h\right\rangle_{\mathcal{E}_h} + \mathcal{N}_{h,\partial\Omega}(\boldsymbol{q}_h, \boldsymbol{w}_h; \boldsymbol{\tau}_h, \boldsymbol{v}_h)
\end{aligned} \tag{9}$$

holds for all $\mathbb{y}_h := (\boldsymbol{\tau}_h, \boldsymbol{v}_h, \boldsymbol{\mu}_h) \in \mathbb{X}_h$. Here we have used the abbreviation $(\cdot,\cdot)$ and $\langle\cdot,\cdot\rangle$ to distinguish between element- and edge-oriented inner products, and the jump operator for the vector valued function is defined as $[\![\boldsymbol{v}]\!] := \boldsymbol{v}^+\cdot\boldsymbol{n} + \boldsymbol{v}^-\cdot\boldsymbol{n}^-$ where the signs $\pm$ correspond to elements κ^+ and κ^- separated by edge e. Here, the analytical flux functions are replaced by the numerical fluxes. The choice of the convective and diffusive numerical fluxes, which we will use, corresponds to Lax-Friedrichs flux and LDG flux, respectively, i.e.,

$$\widehat{\boldsymbol{f}}_c(\boldsymbol{\lambda}_h, \boldsymbol{w}_h) = \boldsymbol{f}_c(\boldsymbol{\lambda}_h)\cdot\boldsymbol{n} - \alpha_c\left(\boldsymbol{\lambda_h} - \boldsymbol{w}_h\right), \tag{10}$$

$$\widehat{\boldsymbol{f}}_v(\boldsymbol{\lambda}_h, \boldsymbol{w}_h, \boldsymbol{q}_h) = \boldsymbol{f}_v(\boldsymbol{\lambda}_h, \boldsymbol{q}_h)\cdot\boldsymbol{n} + \alpha_v\left(\boldsymbol{\lambda_h} - \boldsymbol{w}_h\right). \tag{11}$$

where we assume α_c and α_v to be constant scalar values.

The boundary conditions are incorporated into the scheme by evaluating the exact flux functions with a suitably chosen vector of conservative variables at the domain boundary $\boldsymbol{w}_{h,\partial\Omega}$ and its gradient $\boldsymbol{q}_{h,\partial\Omega}$ depending on a particular type of the boundary condition,

$$\begin{aligned}
\mathcal{N}_{h,\partial\Omega}(\boldsymbol{q}_h, \boldsymbol{w}_h; \boldsymbol{\tau}_h, \boldsymbol{v}_h) &= \left\langle \boldsymbol{w}_{h,\partial\Omega}(\boldsymbol{w}_h), \boldsymbol{\tau}_h\cdot\boldsymbol{n}\right\rangle_{\mathcal{E}_h^\partial} \\
&+ \left\langle \left(\boldsymbol{f}_c(\boldsymbol{w}_{h,\partial\Omega}(\boldsymbol{w}_h)) - \boldsymbol{f}_v(\boldsymbol{w}_{h,\partial\Omega}(\boldsymbol{w}_h), \boldsymbol{q}_{h,\partial\Omega}(\boldsymbol{q}_h))\right)\cdot\boldsymbol{n}, \boldsymbol{v}\right\rangle_{\mathcal{E}_h^\partial}.
\end{aligned} \tag{12}$$

The presence of an unsteady term in the semi-discretized scheme (9) gives rise to a system of differential-algebraic equations (DAEs) of index 1. Hence the classical approach using the method of lines with some explicit time integration scheme cannot be used, and one has to choose an appropriate implicit method for solving DAEs. If we split the discretization of the term with the time derivative and the remaining part, we can write the formulation as follows: Find $\mathbb{x}_h \in \mathbb{X}_h$ such that

$$\mathcal{T}(\partial_t\mathbb{x}_h; \mathbb{y}_h) + \mathcal{N}(\mathbb{x}_h; \mathbb{y}_h) = 0 \quad \forall \mathbb{y}_h \in \mathbb{X}_h \tag{13}$$

where $\mathcal{T}(\partial_t\mathbb{x}_h; \mathbb{y}_h) = (\partial_t\boldsymbol{w}_h, \boldsymbol{v}_h)_{\mathcal{T}_h}$. In this work, we consider the Diagonally Implicit Runge-Kutta (DIRK) methods for the discretization of the time derivative. Applying a general s-stage DIRK method to formulation (13) results in

$$\frac{1}{a_{ii}\Delta t^n}\mathcal{T}(\mathbb{x}_h^{n,i}; \mathbb{y}_h) + \mathcal{N}(\mathbb{x}_h^{n,i}; \mathbb{y}_h) = \frac{1}{a_{ii}\Delta t^n}\mathcal{T}(\mathbb{x}_h^{n}; \mathbb{y}_h) - \sum_{j=1}^{i-1}\frac{a_{ij}}{a_{ii}}\mathcal{N}(\mathbb{x}_h^{n,j}; \mathbb{y}_h) \quad \forall \mathbb{y}_h \in \mathbb{X}_h, \tag{14}$$

which has to be solved for the intermediate solutions in each stage $i = 1, \dots, s$. This step can also be seen as an implicit Euler method with modified right-hand side and coefficients a_{ij}, $i, j = 1, \dots, s$ given by a particular DIRK method. The solution at the new time level t^{n+1} is then obtained by

$$\mathcal{T}(\mathbb{x}_h^{n+1}; \mathbb{y}_h) = \mathcal{T}(\mathbb{x}_h^{n}; \mathbb{y}_h) - \Delta t^n \sum_{i=1}^{s} b_i \mathcal{N}(\mathbb{x}_h^{n,i}; \mathbb{y}_h) \quad \forall \mathbb{y}_h \in \mathbb{X}_h. \tag{15}$$

We consider three types of DIRK methods all being so called stiffly accurate [14], meaning that the coefficients a_{ij} and b_i in (14) and (15), respectively, satisfy $a_{sj} = b_j$, $j = 1, \dots, s$, and hence the numerical solution at the time step $n+1$ is identical to the intermediate solution of the last stage such that the final step (15) of the scheme is avoided. In particular, we use the two-stage second-order DIRK(2,2) method of Alexander [15], the three-stage third-order DIRK(3,3) of Cash [16], and the five-stage fourth-order DIRK(5,4) method of Hairer and Wanner [14]. The resulting system of nonlinear equations in each stage of a DIRK method (14) is solved by a damped Newton-Raphson method.

IV. Mesh Adaptation

We present a brief overview of metric-based mesh adaptation as particular concepts are exploited in our anisotropic mesh adaptation algorithm. The fundamental concept is the mesh-metric duality and the continuous mesh model introduced by Loseille and Alauzet [17, 18]. Given a triangulation $\mathcal{T}_h = \{\kappa\}$ of the computational domain $\Omega \in \mathbb{R}^2$, each nondegenerate simplex element can be characterized by symmetric positive definite matrix

$$\boldsymbol{M} = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \tag{16}$$

such that each of the edges $\boldsymbol{e}_i$, $i = 1, 2, 3$, of a triangular element has a constant lenght C under the norm induced by the metric

$$\|\boldsymbol{e}_i\|_{\boldsymbol{M}} = \sqrt{\boldsymbol{e}_i^T \boldsymbol{M} \boldsymbol{e}_i} = C, \quad i = 1, 2, 3. \tag{17}$$

Equation (17) now represents a system of linear equations for the elements a, b, and c of the matrix $\boldsymbol{M}$. The mesh element can be inscribed into an ellipse with its origin located at the element centroid, h_1 and h_2 being its principal axes and θ its orientation w.r.t. Cartesian coordinate system. These quantities can be extracted from the spectral decomposition of the matrix, i.e. $\boldsymbol{M} = \boldsymbol{Q}^T \boldsymbol{\Lambda} \boldsymbol{Q}$ where

$$\boldsymbol{Q} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \boldsymbol{\Lambda} = \begin{pmatrix} \frac{1}{h_1^2} & 0 \\ 0 & \frac{1}{h_2^2} \end{pmatrix}. \tag{18}$$

One can also define the aspect ratio $\beta := h_2/h_1$ and the local density $d := 1/(h_1 h_2)$, which is inversely related to the area of the element $|\kappa|$. The tuple (d, β, θ) now completely defines the metric $\boldsymbol{M}$, and hence the mesh element $\kappa \in \mathcal{T}_h$. Quantities β and θ can be denoted as the anisotropy of the triangle.

Our adaptation methodology is based on evaluation of the indicated metric for each element of the current mesh and generation of an optimized metric, which is obtained using the solution error estimate, corresponding to the desired mesh element.

The metric described above is discrete in the sense that it is defined for each element. Using a suitable interpolation, one may generate a continuous metric field $\mathcal{M}(\boldsymbol{x})$, $\boldsymbol{x} \in \Omega$, which can then be used in a metric-based mesh generator to produce the desired mesh. In the present work, we use the `BAMG` mesh generator of Hecht [19]. The length of the edge $\boldsymbol{e} = \boldsymbol{x}_2 - \boldsymbol{x}_1$ in the continuous setting is computed using a straight line parametrization,

$$\|\boldsymbol{e}\|_{\mathcal{M}} = \int_0^1 \sqrt{\boldsymbol{e}^T \mathcal{M}(\boldsymbol{x}_1 + s\boldsymbol{e}) \boldsymbol{e}} \, \mathrm{d}s. \tag{19}$$

Edges of the resulting triangulation will then satisfy $\|\boldsymbol{e}\|_{\mathcal{M}} \approx C$ where C is the constant in Eq. (17). The generated mesh can be viewed as a discrete approximation of the continuous mesh, which is represented by the continuous metric field $\mathcal{M}(\boldsymbol{x})$.

The aim of the anisotropic mesh adaptation is therefore to optimize both the size and the anisotropy of the mesh elements in such a way that the resulting mesh will have improved approximation properties. In our case, the goal is to obtain triangle that gives the smallest interpolation error into the piecewise polynomial approximation space (7) in the L^q-norm. For a sufficiently smooth function $u(\boldsymbol{x})$, $\boldsymbol{x} \equiv (x_1, x_2)^T \in \Omega$, the local error model based on the work of Dolejší [20] is making use of the remainder term in the Taylor series of order p centered at $\boldsymbol{x}$. For this purpose, let us now define the directional derivative of k-th order of function u as

$$u^{k,\phi}(\boldsymbol{x}) = \sum_{\ell=0}^{k} \binom{k}{\ell} \frac{\partial^k u(\boldsymbol{x})}{\partial x_1^\ell \partial x_2^{k-\ell}} (\cos\phi)^\ell (\sin\phi)^{k-\ell}, \quad \phi \in [0, 2\pi). \tag{20}$$

The error model is then given by

$$|e_{\boldsymbol{x},p}(\boldsymbol{y})| = \frac{1}{(p+1)!} \left| u^{(p+1,\phi)}(\boldsymbol{x}) \right| |\boldsymbol{y} - \boldsymbol{x}|^{p+1}, \quad \forall \boldsymbol{y} \in \Omega. \tag{21}$$

Following [20], one can obtain a bound for the interpolation error in terms of three parameters (A_p, ρ_p, ϕ_p) that are related to the anisotropy of the element (β, θ). Let ϕ_p corresponds to the angle where the directional derivative is reaching its maximum,

$$\phi_p \equiv \phi_p(\boldsymbol{x}) := \underset{\phi \in [0,2\pi)}{\operatorname{argmax}} \left| u^{(p+1,\phi)}(\boldsymbol{x}) \right|, \tag{22}$$

and let $\phi_p^\perp$ identify the direction orthogonal to ϕ_p. Then, A_p is the maximum of the scaled $(p+1)$-st directional derivative and ρ_p is the ratio of the maximum derivative and the derivative along its perpendicular direction given by ϕ_p, i.e.,

$$A_p \equiv A_p(\boldsymbol{x}) := \frac{1}{(p+1)!} \left| u^{(p+1,\phi_p)}(\boldsymbol{x}) \right| \qquad \text{and} \qquad \rho_p \equiv \rho_p(\boldsymbol{x}) := \frac{A_p(\boldsymbol{x})}{A_p^\perp(\boldsymbol{x})} \tag{23}$$

where $A_p^\perp := |u^{(p+1,\phi_p^\perp)}(\boldsymbol{x})|$. For a triangle $\kappa \in \mathcal{T}_h$ centered at $\boldsymbol{x}_\kappa$, the optimal interpolation error becomes

$$\|e_{\boldsymbol{x}_\kappa,p}\|_{L^q(\kappa)} \le e_d(\boldsymbol{x}_\kappa)^q d^{-1}(\boldsymbol{x}_\kappa) \qquad \text{with} \qquad e_d(\boldsymbol{x}) := \left(\frac{2}{q(p+1)+2} \right)^{\frac{1}{q}} A_p(\boldsymbol{x}) \rho_p(\boldsymbol{x})^{-1/2} d(\boldsymbol{x})^{-\frac{p+1}{2}}. \tag{24}$$

The optimal value of the interpolation error is achieved for $\beta = \rho_p^{1/(p+1)}$ and $\theta = \phi_p - \pi/2$. We refer to [20] for details. The anisotropy parameters (β, θ) are thus eliminated and replaced by the higher derivatives of u with the help of parameters (A_p, ρ_p). With fixed optimal anisotropy, the corresponding error estimate is then valid for a family of ellipses where the local density d, hence the size of the triangle, is now a free parameter that can be chosen based on some appropriate optimization strategy.

Suppose that we have a mesh composed of elements with arbitrary size distribution but each having locally optimal anisotropy. Then starting from (24) the following holds asymptotically,

$$\|e_{\boldsymbol{x}_\kappa,p}\|_{L^q(\Omega)} \le \sum_{\kappa \in \mathcal{T}_h} e_d(\boldsymbol{x}_\kappa)^q d^{-1}(\boldsymbol{x}_\kappa) \propto \sum_{\kappa \in \mathcal{T}_h} e_d(\boldsymbol{x}_\kappa)^q |\kappa| \to \int_\Omega e_d(\boldsymbol{x})^q \, \mathrm{d}\boldsymbol{x} \qquad (h \to 0). \tag{25}$$

With a continuous error model one may now apply analytic optimization techniques to obtain the optimal size distribution given the number of mesh elements as a constraint. Hence we consider the following constrained optimization problem:

$$d^* = \underset{d}{\operatorname{argmin}} \int_\Omega e_d(\boldsymbol{x})^q \, \mathrm{d}\boldsymbol{x} \qquad \text{such that} \qquad \int_\Omega d(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x} = N \tag{26}$$

where N is the mesh complexity related to the number of mesh elements. Straightforward application of calculus of variations leads the following result

$$d^*(\boldsymbol{x}) = K \left(A_p(\boldsymbol{x}) \rho_p(\boldsymbol{x})^{-1/2} \right)^{\frac{2q}{q(p+1)+2}} \qquad \text{with} \qquad K = \frac{N}{\int_\Omega \left(A_p(\boldsymbol{x}) \rho_p(\boldsymbol{x})^{-1/2} \right)^{\frac{2q}{q(p+1)+2}}}. \tag{27}$$

The details of the outlined algorithm can be found in earlier work [8]. The optimization described above has also been extended to goal-oriented adaptation based on the solution of an adjoint problem, see [21, 22].

V. Mesh-to-Mesh Interpolation

We have seen in Sec. III that in order to solve Eq. (14) for the intermediate solutions in each stage of the DIRK method, a knowledge of the solution from the previous time step is needed. Hence the use of anisotropic mesh refinement in unsteady simulations relies on the solution transfer between the previous and the current mesh. Furthermore, this interpolation step needs special care since we would like to avoid the disruption of the design order of convergence of the HDG method and/or particular DIRK scheme. Another important property whose absence can have fatal consequences for the evolution of numerical solution in time is the conservation of the interpolation method.

The solution transfer algorithm used in this study is based on the work of Farrell [10, 11], where Galerkin projection is exploited for the purpose of interpolation between two meshes. The Galerkin projection represents the optimally accurate projection in the L^2-norm, and it is especially beneficial for the piecewise discontinuous fields arising in the DG discretizations as the interpolation can be done locally on each element. Moreover, from the fact that the resulting field minimizes the interpolation error in L^2-norm, the conservation follows naturally.

Consider field w_{old}, which can also be a component of the vector of conservative variables appearing in the balance law (1), defined on mesh $\mathcal{T}_{h,\text{old}}$, resulting from the n-th time step of simulation of an unsteady problem. After a mesh adaptation step, a new adapted triangulation $\mathcal{T}_{h,\text{new}}$ based on the solution w_{old} is obtained. Let each element κ of the mesh $\mathcal{T}_{h,\text{old}}$ have N^{κ}_{old} degrees of freedom corresponding to $\varphi^{(i)}_{\text{old}}$, $i = 1, \dots, N^{\kappa}_{\text{old}}$, basis functions. Similarly, for the case of the adapted mesh, let N^{κ}_{new} be the degrees of freedom of each element $\kappa \in \mathcal{T}_{h,\text{new}}$ with $\varphi^{(i)}_{\text{new}}$, $i = 1, \dots, N^{\kappa}_{\text{new}}$, basis functions. Both fields w_{old} and w_{new} can then be represented on element κ by the linear expansion

$$w_{\text{old}} = \sum_{i=1}^{N^{\kappa}_{\text{old}}} W^{(i)}_{\text{old}} \varphi^{(i)}_{\text{old}},\ \kappa \in \mathcal{T}_{h,\text{old}} \qquad \text{and} \qquad w_{\text{new}} = \sum_{i=1}^{N^{\kappa}_{\text{new}}} W^{(i)}_{\text{new}} \varphi^{(i)}_{\text{new}},\ \kappa \in \mathcal{T}_{h,\text{new}}. \tag{28}$$

The aim is to obtain an interpolant w_{new} on each element of $\mathcal{T}_{h,\text{new}}$, which is optimal in the L^2-norm, i.e.,

$$\|w_{\text{old}} - w_{\text{new}}\|_{L^2(\kappa)} = \min_{w \in V^{\kappa}_{\text{new}}} \|w_{\text{old}} - w\|_{L^2(\kappa)} \tag{29}$$

where $V^{\kappa}_{\text{new}} = \text{span}\{\varphi^{(i)}_{\text{new}}, i = 1, \dots, N^{\kappa}_{\text{new}}\}$. Ideally, we would like to obtain the interpolated solution w_{new} such that $\forall \boldsymbol{x} \in \Omega : w_{\text{old}}(\boldsymbol{x}) = w_{\text{new}}(\boldsymbol{x})$, in which case the left-hand side of (29) is zero. Naturally, this is not possible with $w_{\text{new}} \in V^{\kappa}_{\text{new}}$. Instead, the equality then can be satisfied in the weak sense by

$$\int_{\kappa} w_{\text{old}} \varphi^{(i)}_{\text{new}}\, \mathrm{d}\boldsymbol{x} = \int_{\kappa} w_{\text{new}} \varphi^{(i)}_{\text{new}}\, \mathrm{d}\boldsymbol{x}, \quad i = 1, \dots, N^{\kappa}_{\text{new}},\ \kappa \in \mathcal{T}_{h,\text{new}}. \tag{30}$$

Since the basis functions $\varphi^{(i)}_{\text{old}}$ in (28) are discontinuous piecewise polynomials on each element of $\mathcal{T}_{h,\text{old}}$, integrating the product of different basis functions numerically by evaluating $\varphi^{(i)}_{\text{old}}$ at each quadrature point of $\kappa \in \mathcal{T}_{h,\text{new}}$ would result in inexact integration, and hence the loss of accuracy and conservativity, as the quadrature rules are exact for a specified polynomial order.

In order to evaluate the corresponding integral, we will utilize the concept of a supermesh [10]. The basis functions $\varphi^{(i)}_{\text{new}}$ are polynomials over $\kappa \in \mathcal{T}_{h,\text{new}}$, and hence the idea is to split the element κ into simplical intersections with the elements of $\mathcal{T}_{h,\text{old}}$ and evaluate the integral as a sum of integrals over these intersections. Therefore, the supermesh is defined as the mesh of intersection polygons of the old and new elements of $\mathcal{T}_{h,\text{old}}$ and $\mathcal{T}_{h,\text{new}}$, respectively, see Fig. 1. The local supermesh construction on element $\kappa \in \mathcal{T}_{h,\text{new}}$ consists of the following steps:

1) Find element $\kappa_1 \in \mathcal{T}_{h,\text{old}}$ intersecting element κ by a brute force method requiring that the centroid of κ lies inside κ_1.
2) The remaining intersecting elements $\kappa_i \in \mathcal{T}_{h,\text{old}}$, $i = 2, \dots, N_I$, are found by the advancing front algorithm [11, 23], which is illustrated in Fig. 2.
3) For each intersecting element $\kappa_i \in \mathcal{T}_{h,\text{old}}$, $i = 1, \dots, N_I$, the nodes forming an intersection polygon are found by searching for an intersection of two lines formed by the faces of elements κ_i and κ that are considered in turn.
4) Once the intersection polygons are identified, they are further split in order to obtain a supermesh $\mathcal{T}_{\kappa} = \{\kappa_{\ell}, \ell = 1, \dots, N_{\kappa}\}$ composed of triangles only.

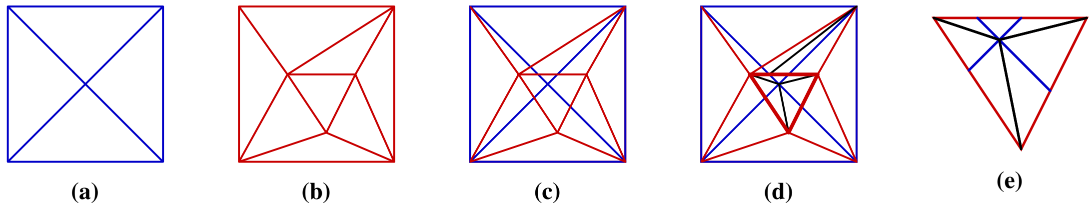


Fig. 1 Illustration of a supermesh for a square domain. (a) Initial mesh $\mathcal{T}_{h,\text{old}}$. (b) Adapted mesh $\mathcal{T}_{h,\text{new}}$. (c) Intersection of the meshes $\mathcal{T}_{h,\text{old}} \cup \mathcal{T}_{h,\text{new}}$ including nonsimplicial intersection elements. (d) The global supermesh composed of simplices. (e) Local supermesh of the middle element of the new mesh $\mathcal{T}_{h,\text{new}}$.

Let $\mathcal{T}_\kappa$ be the local supermesh of element $\kappa \in \mathcal{T}_{h,\text{new}}$ having N_κ elements. Now replacing w_{old} and w_{new} in (30) by their finite element representations (28) and taking the supermesh elements into account yield

$$\sum_{\ell=1}^{N_\kappa} \int_{\kappa_\ell \in \mathcal{T}_\kappa} \sum_{j=1}^{N_{\text{old}}^\kappa} W_{\text{old}}^{(j)} \varphi_{\text{old}}^{(j)} \varphi_{\text{new}}^{(i)} \,\mathrm{d}\boldsymbol{x} = \sum_{\ell=1}^{N_\kappa} \int_{\kappa_\ell \in \mathcal{T}_\kappa} \sum_{k=1}^{N_{\text{new}}^\kappa} W_{\text{new}}^{(k)} \varphi_{\text{new}}^{(k)} \varphi_{\text{new}}^{(i)} \,\mathrm{d}\boldsymbol{x}, \quad i = 1, \dots, N_{\text{new}}^\kappa. \tag{31}$$

This representation gives rise to a linear system for the vector of expansion coefficients $\boldsymbol{W}_{\text{new}}$ on each $\kappa \in \mathcal{T}_{h,\text{new}}$ in the form

$$\boldsymbol{M}_{\text{new}} \boldsymbol{W}_{\text{new}} = \boldsymbol{M}_{\text{old,new}} \boldsymbol{W}_{\text{old}} \tag{32}$$

where $\boldsymbol{M}_{\text{new}}$ is the mass matrix defined with the set of basis functions φ_{new} and $\boldsymbol{M}_{\text{old,new}}$ is the mixed mass matrix of both sets of basis functions φ_{new} and φ_{old} with components

$$(\boldsymbol{M}_{\text{new}})_{ij} = \sum_{\ell=1}^{N_\kappa} \int_{\kappa_\ell \in \mathcal{T}_\kappa} \varphi_{\text{new}}^{(i)} \varphi_{\text{new}}^{(j)} \,\mathrm{d}\boldsymbol{x}, \quad i, j = 1, \dots, N_{\text{new}}^\kappa, \tag{33}$$

$$(\boldsymbol{M}_{\text{old,new}})_{ij} = \sum_{\ell=1}^{N_\kappa} \int_{\kappa_\ell \in \mathcal{T}_\kappa} \varphi_{\text{new}}^{(i)} \varphi_{\text{old}}^{(j)} \,\mathrm{d}\boldsymbol{x}, \quad i = 1, \dots, N_{\text{new}}^\kappa, j = 1, \dots, N_{\text{old}}^\kappa. \tag{34}$$

Note that there is no need to explicitly store the mixed mass matrix as in order to assemble the right-hand side of (32) only the product $\boldsymbol{M}_{\text{old,new}} \boldsymbol{W}_{\text{old}}$ is required.

A bounded Galerkin projection is proposed in [10] in order to preserve minima and maxima of the field defined on the old mesh. However, this boundeness is restricted to linear basis functions only. At this stage, we have not incorporated any bounding procedure into the algorithm.

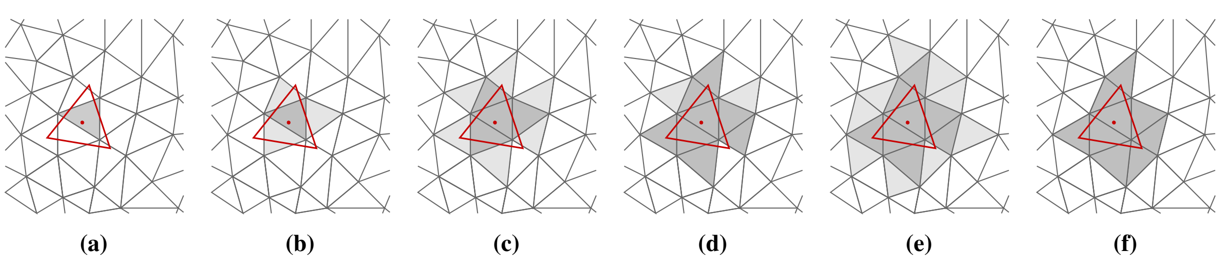


Fig. 2 Illustration of element intersection searching via the advancing front algorithm. The gray background mesh represents the mesh $\mathcal{T}_{h,\text{old}}$ and the element $\kappa \in \mathcal{T}_{h,\text{new}}$ is shown in red. (a) Element $\kappa_1 \in \mathcal{T}_{h,\text{old}}$ containing the centroid of κ is found. (b) The neighbor elements of κ_1 are identified and tested whether they are intersecting with κ. (c) - (e) This algorithm is applied for successive neighbors of intersecting elements until no new intersections are found. (f) The resulting set of intersecting elements.

VI. Numerical Results

A. Repeated Interpolation

The first test case does not involve any solution of a conservation law (1). Instead, a given function is interpolated several times between structured and unstructured meshes defined on the domain $\Omega = [-1, 1]^2$ and having the same characteristic element size h but different topology, see Fig. 3. The purpose of this test case is to investigate the observed order of convergence of the Galerkin projection for the field represented by polynomials of degrees $p = 1, \dots, 4$ on each element.

We consider two functions, namely a sine wave and a smooth approximation to a step function,

$$w_1 = \sin(\pi x) \sin(\pi y), \tag{35}$$

$$w_2 = \frac{10x}{\sqrt{1 + (10x)^2}} + \frac{10y}{\sqrt{1 + (10y)^2}}. \tag{36}$$

Both of these functions are initialized on the structured mesh and the field is then interpolated onto the unstructured one, followed by interpolation back to the structured mesh. This interpolation procedure is repeated 10 times.

In order to compare the properties of Galerkin projection with some other approach, we assume the solution existing on the old mesh is simply evaluated on the points of the new mesh. This approach is computationally less demanding since no supermesh construction is needed, but, as we will see, this method is generally not accurate nor conservative.

The convergence results for both functions are shown in Figs. 4 and 5. As expected, the observed order of convergence is $O(h^{p+1})$ for the case of Galerkin projection. The solution evaluation approach performs slightly worse for the smooth step function w_2. However, if we compare the integral of the final interpolant over the domain Ω, we see that only the Galerkin projection results in conservative interpolation method. Note that the exact integral is zero for both functions (35)-(36).

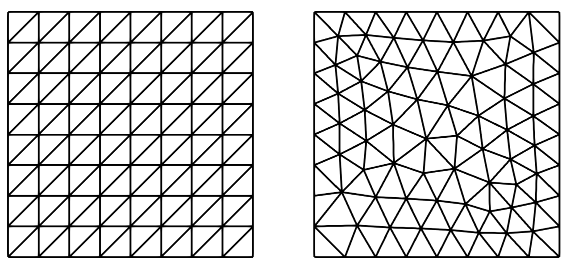


Fig. 3 Structured and unstructured meshes with $h = 0.25$ used for the interpolation test case

B. Rotating Gaussian

Next, we consider the 2D scalar linear convection-diffusion equation of the form

$$\frac{\partial w}{\partial t} + \nabla \cdot (\boldsymbol{u} w) = \varepsilon \nabla^2 w, \quad \boldsymbol{x} \in \Omega. \tag{37}$$

The idea is to couple the mesh adaptation algorithm with the unsteady solution procedure such that the mesh is adapted in each time step once the current solution is known. Before the mesh is being used within the next time step, the solution transfer approach based on the Galerkin projection is applied as described in Sec. V.

The particular test case involving the rotational transport of a Gaussian pulse inside the domain $\Omega = [-1, 1]^2$ has been studied in [24]. With the velocity field set to $\boldsymbol{u}(x, y) = (-4y, 4x)$, the exact solution for this problem is given by

$$w(x, y, t) = \frac{2\sigma^2}{2\sigma^2 + 4\varepsilon t} \exp\left(-\frac{(\widehat{x} - x_c)^2 + (\widehat{y} - y_c)^2}{2\sigma^2 + 4\varepsilon t}\right) \tag{38}$$

where

$$\widehat{x} = x\cos(4t) + y\sin(4t), \qquad \widehat{y} = -x\sin(4t) + y\cos(4t). \tag{39}$$

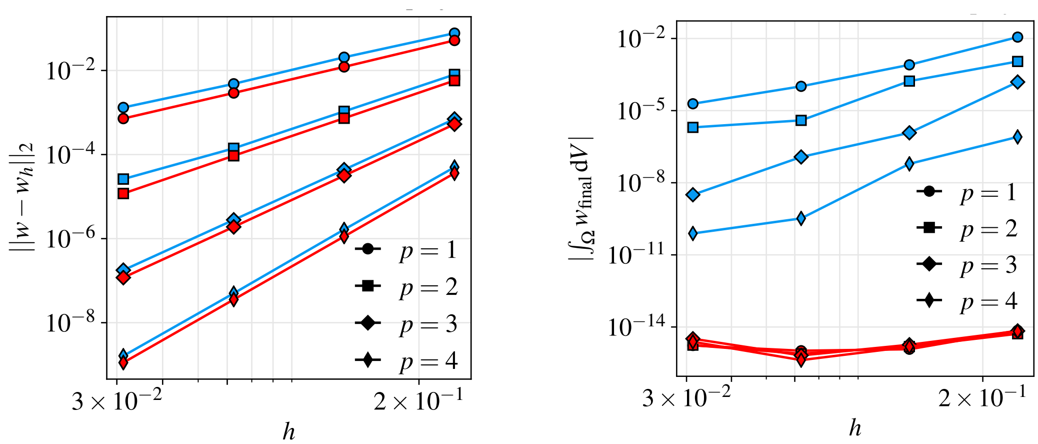


Fig. 4 Convergence results for the L^2-norm error of function w_1 (left) and its integral over the domain Ω (right) as a function of the characteristic mesh size h, old solution evaluation (—), Galerkin projection (—).

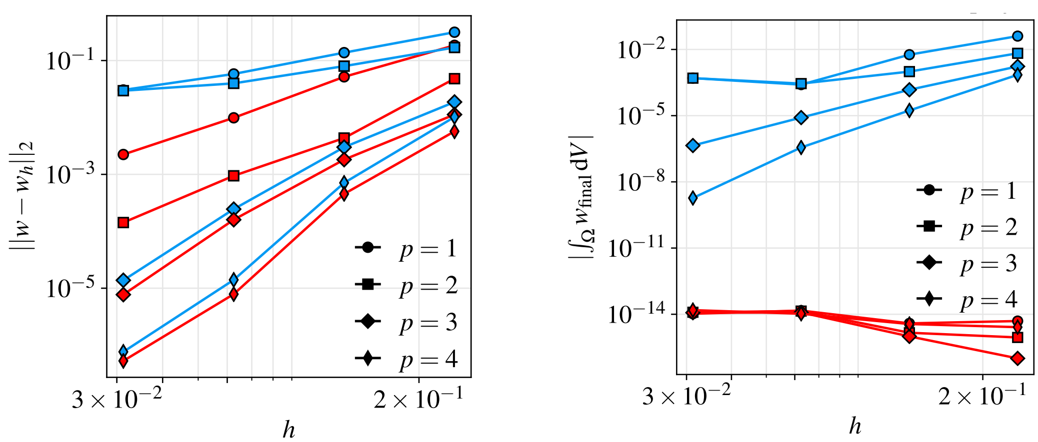


Fig. 5 Convergence results for the L^2-norm error of function w_2 (left) and its integral over the domain Ω (right) as a function of the characteristic mesh size h, old solution evaluation (—), Galerkin projection (—).

The initial center of the pulse is chosen as $(x_c, y_c) = (-0.1, 0)$, the standard deviation of the Gaussian distribution is set to $\sigma = 0.1$, and the diffusivity constant is set to $\varepsilon = 0.001$. Dirichlet boundary conditions taken from the exact solution are assumed at $\partial\Omega$. We will be interested in the solution at the final time $T = \pi/4$ that corresponds to one-half rotation of the pulse in counterclockwise direction.

We use the DIRK(5,4) method and perform the unsteady simulation with fixed number of elements N_e for every mesh adaptation. The test case is run successively with $N_e = 20, 80, 320, 1280$. In order to observe a spatial solution error, the time step size Δt is set to $T/(0.5N_e)$. Based on the numerical experiments, this choice is sufficient for the temporal error to be small enough such that the overall error is governed strictly by the spatial discretization error.

The solution together with the adapted mesh at the final time is depicted in Fig. 6 for the case $N_e = 1280$. The convergence history is tabulated in Tab. 1. One can see that numerical solutions of polynomial degree $p = 1, \ldots, 4$ converge with order $p + 1$, and hence the Galerkin projection together with the HDG discretization leads to optimal convergence rate as in the situation with a fixed mesh. To assess the conservative properties of the interpolation, an integral of the solution over domain Ω is plotted in Fig. 7 as a function of time for the case of the mesh with the largest number of elements. It can be seen that even for the piecewise linear solution, the overall integral deviates from the exact one approximately by 2×10^{-6}.

Table 1 History of convergence in terms of the error $\|w - w_h\|_2$

Degree	Mesh N_e	Error	Order	Degree	Mesh N_e	Error	Order
1	20	$1.036 \cdot 10^{-1}$	-	3	20	$1.045 \cdot 10^{-2}$	-
	80	$1.814 \cdot 10^{-2}$	2.51		80	$4.883 \cdot 10^{-4}$	4.42
	320	$2.941 \cdot 10^{-3}$	2.63		320	$2.082 \cdot 10^{-5}$	4.55
	1280	$6.522 \cdot 10^{-4}$	2.17		1280	$7.810 \cdot 10^{-7}$	4.74
2	20	$4.907 \cdot 10^{-2}$	-	4	20	$9.480 \cdot 10^{-3}$	-
	80	$2.242 \cdot 10^{-3}$	4.45		80	$1.187 \cdot 10^{-4}$	6.32
	320	$1.901 \cdot 10^{-4}$	3.56		320	$3.498 \cdot 10^{-6}$	5.09
	1280	$1.934 \cdot 10^{-5}$	3.30		1280	$3.166 \cdot 10^{-8}$	6.79

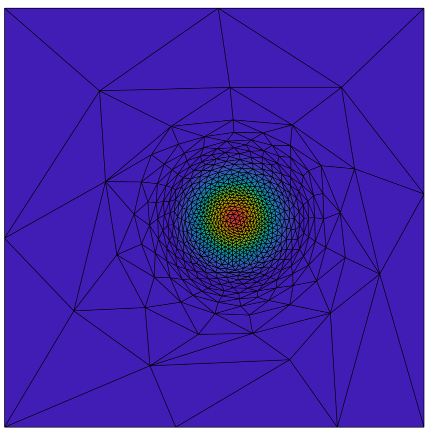


Fig. 6 Adapted mesh at the final time $T = \pi/4$ with $N_e = 1280$

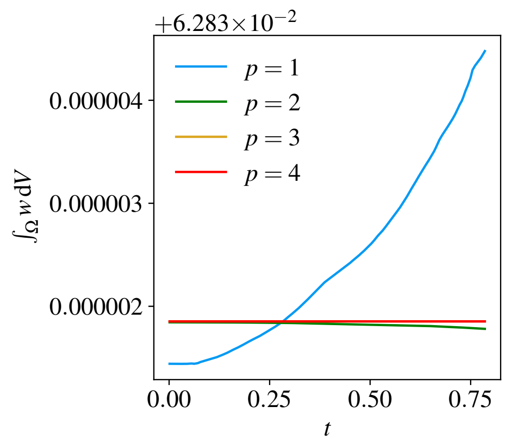


Fig. 7 Evolution of the solution integral over the time for $p = 1, \ldots, 4$ and $N_e = 1280$

C. Pure Advection of Discontinuous Scalar Field

In order to assess the properties of the presented overall numerical method for the situations where the solution is discontinuous, we first consider conservation law (37) with zero diffusivity coefficient, i.e. the linear advection equation in the form

$$\frac{\partial w}{\partial t} + \nabla \cdot (\boldsymbol{u} w) = 0, \quad \boldsymbol{x} \in \Omega. \tag{40}$$

We consider the test case introduced by Rudman [25], where the initial condition is given by a circular region of unit value and radius $R = \pi/5$ centered at $(x_c, y_c) = (\pi/2, (\pi + 1)/5)$, i.e.

$$w(x, y, 0) = \begin{cases} 1 & \text{for } (x - x_c)^2 + (y - y_c)^2 \leq R, \\ 0 & \text{otherwise.} \end{cases} \tag{41}$$

The velocity field is set to $\boldsymbol{u}(x, y) = (\sin(x)\cos(y), -\cos(x)\sin(y))$ representing a shearing flow since $\partial_x u \neq 0$ and $\partial_y v \neq 0$. The initially circular region then deforms during its advection around the domain $\Omega = [0, \pi]^2$. Furthermore, we assume that the sign of the velocity field $\boldsymbol{u}$ is reversed at time $T/2$. In this case, the scalar field distribution at the final time T should be identical to the initial condition. Here, we will assume $T = 20$ and four time step sizes $\Delta t = 0.1, 0.05, 0.025, 0.0125$, corresponding to 200, 400, 800 and 1,600 time steps, respectively. The time step size is kept constant for the whole simulation time, and hence the Courant number is variable depending on the anisotropic adaptation of the mesh. The main purpose of this test case is to demonstrate the utility of the diagonally implicit Runge-Kutta methods DIRK(2,2), DIRK(3,3), and DIRK(5,4) when coupled with the anisotropic mesh adaptation performed after every time step.

From now on, the Galerkin projection is considered as a primary choice for the solution transfer between anisotropically adapted meshes. After each adaptation, the computational mesh is limited to have roughly 2,000 elements. At the same time, we are using $p = 3$ polynomials to represent the numerical solution on each mesh element. The mesh used at the beginning of the computation is depicted in Fig. 8(a). It is obtained by the following procedure:

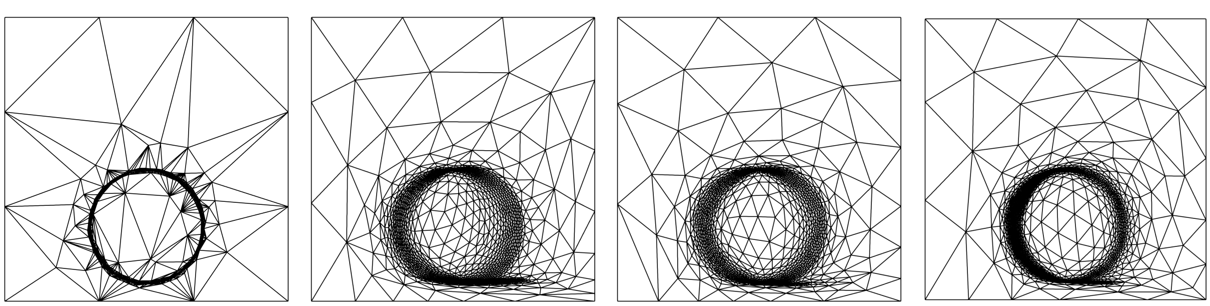
1) Evaluate the initial condition (41) on a uniform triangular mesh $\mathcal{T}_1$ having 8,192 elements.
2) Obtain anisotropically adapted mesh $\mathcal{T}_2$ based on the initial scalar field distribution on mesh $\mathcal{T}_1$.
3) Evaluate the initial condition on mesh $\mathcal{T}_2$ and use this field to obtain anisotropically adapted mesh $\mathcal{T}_3$ (Fig. 8(a)), which is used as the initial mesh for the simulation.

Figs. 8(b) - 8(d) show the mesh at the final time $T = 20$ for the second-order DIRK(2,2) method, third-order DIRK(3,3) method, and fourth-order DIRK(5,4) method, respectively, when the time step $\Delta t = 0.0125$ is used. Ideally, besides the numerical solution, even the final mesh should be identical to the initial one. However, as the discretization errors are gradually introduced into the solution, the circular interface becomes slightly smeared depending on the particular time-integration method. One can see that as the order of the DIRK method is increased, the more diffusive is the distribution of the mesh elements. Nevertheless, upon closer inspection, we find that the anisotropically adapted mesh distribution depends also on the propagation of oscillations in the numerical solution.

Let us now focus on the numerical solution at final time $T = 20$ on the horizontal line $y = (\pi + 1)/5$ going through the center of the initial circular region. The numerical solution oscillates near the discontinuity for all considered DIRK methods. Detailed views of regions close to the left and right discontinuity are depicted in Fig. 9. It can be seen that despite the most clustered mesh elements in the region of the discontinuity for the case of DIRK(5,4) (Fig. 8(d)), the corresponding numerical solution is at the same time the most oscillatory. Similar behaviour can be seen also for the DIRK(2,2) method, where the dispersive properties increase as the time step size is decreased. A slightly different situation occurs for the case of third-order DIRK(3,3) method. Here, as the time step size is decreased, the oscillations do not grow but they are rather damped and clustered near the discontinuity.

As we will see on the next test case, the behavior of the DIRK(3,3) method has a positive impact on the quality of the numerical solution when solving the Euler equations, given a problem with shock waves present in the resulting flow field. A demonstration of this phenomena for the Rudman's test case is shown in Fig. 10, where the DIRK(3,3) is the only method, which does not suffer from propagating oscillations through the numerical solution. Especially for DIRK(2,2), one can see that the oscillations themselves affect the resulting anisotropically adapted mesh distribution.

Let us emphasize that no shock-capturing approach to deal with the discontinuities present in the numerical solution nor bounding procedure during the solution transfer has been applied in this test case. In other words, there is no mechanism to suppress the oscillations arising as the problem under consideration is numerically solved.



(a) Initial **(b) Final, DIRK(2,2)** **(c) Final, DIRK(3,3)** **(d) Final, DIRK(5,4)**

Fig. 8 Initial mesh used at the beginning of the computation (a). Mesh at the final time $T = 20$ resulting from the use of various DIRK methods for integration in time (b) - (d).

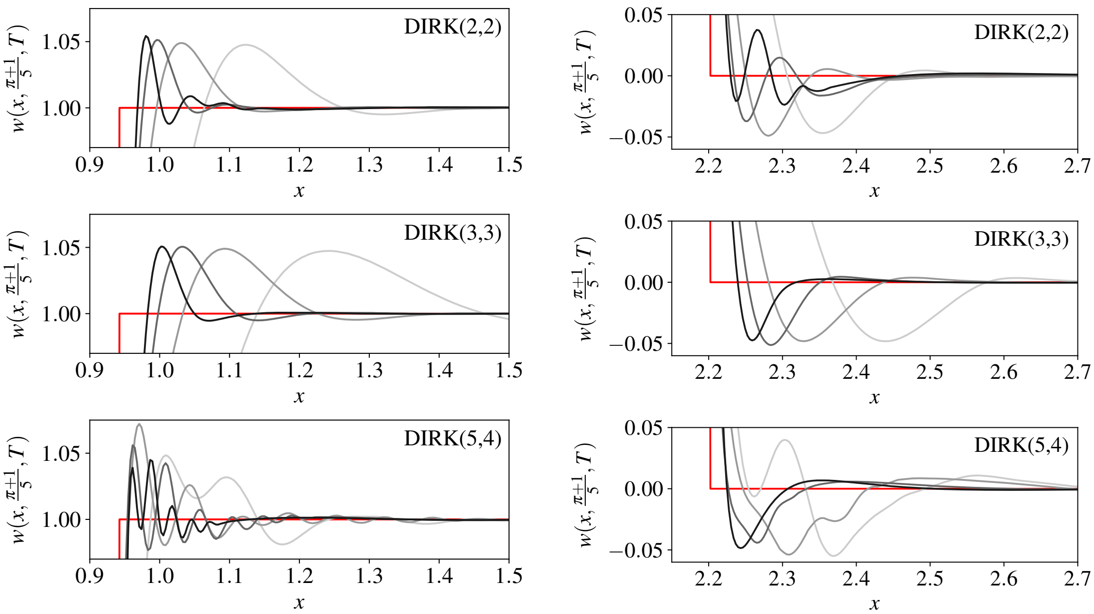
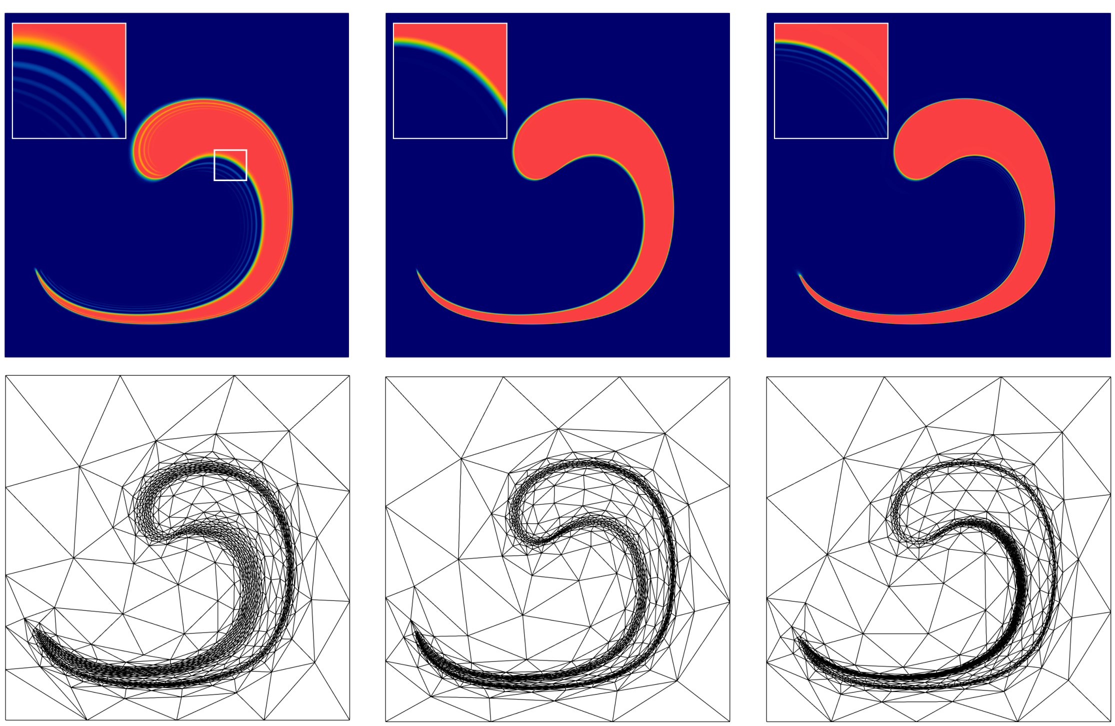


Fig. 9 Close-up view of the solution on the line $y = (\pi + 1)/5$ at the final time $T = 20$ for decreasing time step sizes: $\Delta t = 0.1$ (▬), $\Delta t = 0.05$ (▬), $\Delta t = 0.025$ (▬), $\Delta t = 0.0125$ (▬). The exact solution (▬) corresponds to the initial condition (41).



(a) DIRK(2,2) **(b) DIRK(3,3)** **(c) DIRK(5,4)**

Fig. 10 Numerical solution of the pure advection test case in time $T = 10$ with detailed view near the discontinuity (top) and corresponding anisotropically adapted meshes (bottom) resulting from the use of various DIRK methods for integration in time.

D. Implosion problem

To test the considered DIRK methods for the case of Euler equations, we utilize the implosion problem from [26], which is initially characterized by an overpressured region. A square computational domain $\Omega = [0, 0.3]^2$ is assumed with the initial condition of the form

$$(\rho, u, v, P) = \begin{cases} (1, 0, 0, 1) & \text{for } y \leq x - 0.15, \\ (0.125, 0, 0, 0.14) & \text{otherwise,} \end{cases} \tag{42}$$

see Fig. 11. The initial mesh used for the computation is uniform triangular with 2,048 elements. Due to the topology of the initial mesh, the initial condition is represented exactly. Hence, the initial adaptation of the mesh is not applied in this case. The mesh is then anisotropically adapted based on the Mach number field after every time step and limited to have approximately 3,000 elements.

For this test case, we use $p = 3$ polynomials to represent the solution on each element of the mesh and we seek the numerical solution at the final time $T = 1$ with time step sizes $\Delta t = 0.005, 0.0025$. To resolve the flow field including shocks and to increase the stability of the numerical solution, we employ the shock-capturing approach of Hartmann [12] in which case the resulting artificial viscosity field is defined in an elementwise manner. The work of Barter and Darmofal [27] and some preliminary tests indicate that better results can be achieved with smooth artificial viscosity field. For this purpose, we also utilize the shock-capturing method of Moro et al. [13], which is based on the dilation of the fluid velocity and which has been introduced directly for the HDG method.

First, we consider all three DIRK methods and the Hartmann shock-capturing method with $\Delta t = 0.005$. The contours of density at time $t = 0.7$ obtained by DIRK(2,2), DIRK(3,3), and DIRK(5,4) methods are depicted in Fig. 13 together with the corresponding adapted meshes. It is evident that the solution obtained by DIRK(2,2) and DIRK(5,4) is strongly affected by the oscillations in large part of the computational domain, which arise in the vicinity of shock waves reflected from the boundaries. Accordingly, the element distribution of the anisotropically adapted meshes then cannot realistically follow the motion of the shock waves yielding almost uniform mesh. In contrast, the DIRK(3,3) method is able to damp the oscillations and the topology of the mesh can follow the development of the numerical solution despite the limited number of mesh elements.

Next, we compare the shock-capturing approach of Hartmann and Moro when the DIRK(3,3) time integration method is used with the time step size $\Delta t = 0.0025$. Figs. 14(a) and 14(b) show the respective contours of density at time $t = 0.7$ and the corresponding meshes. In both cases, one can clearly see that the important features of the flow are captured by the topology of the adapted mesh. However, the solution should be symmetric along the major diagonal of the domain, which is clearly not satisfied in the bottom right corner, in the region of numerical instability, for the case of Hartmann's artificial viscosity.

For completeness, the result of the DIRK(5,4) method, where the Moro's artificial viscosity has been applied, is depicted in Fig. 14(c). Again, it can be seen that most of the elements lie in the region of oscillations which in turn results in poorly resolved flow field compared to the numerical solution obtained by DIRK(3,3). The artificial viscosity fields are shown in Fig. 12. It is evident that the Moro's shock sensor detects the oscillations but it is unable to suppress them.

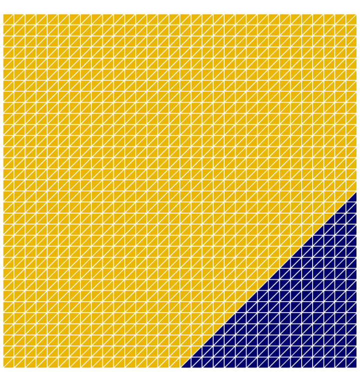
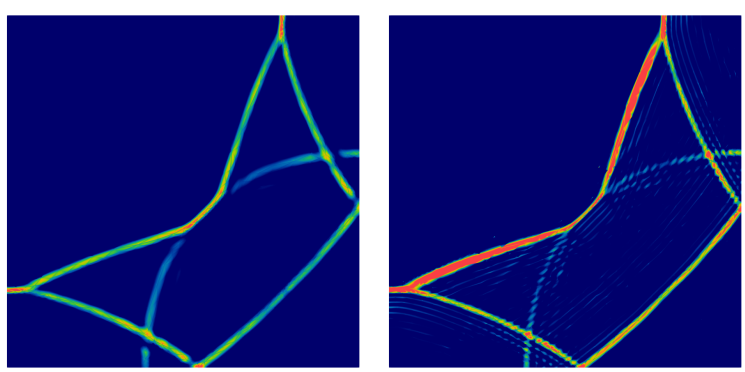
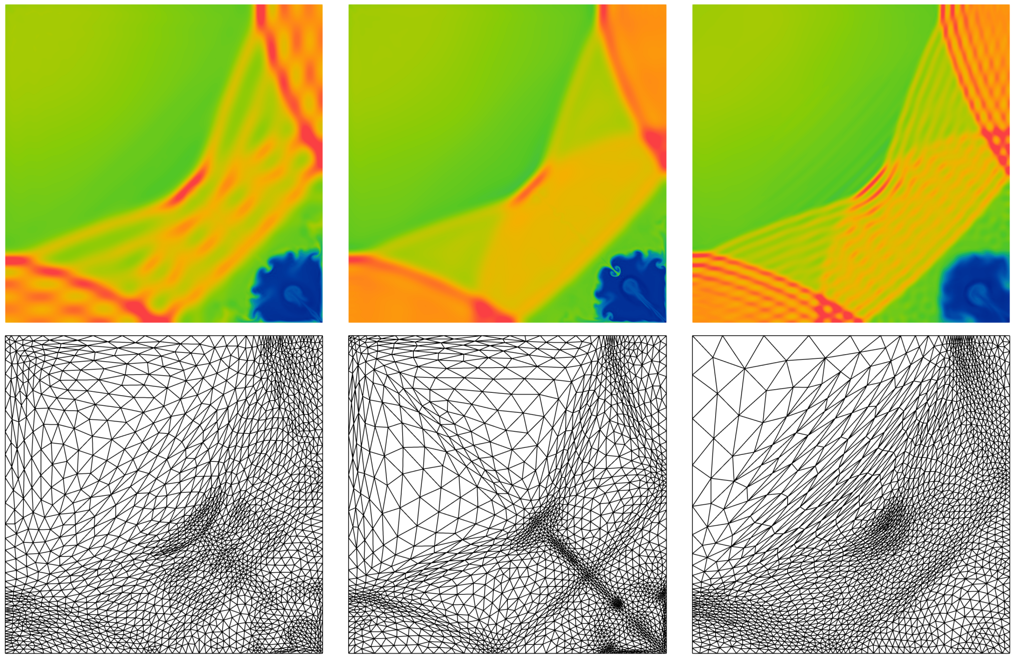


Fig. 11 Initial condition for the implosion problem and the initial mesh.



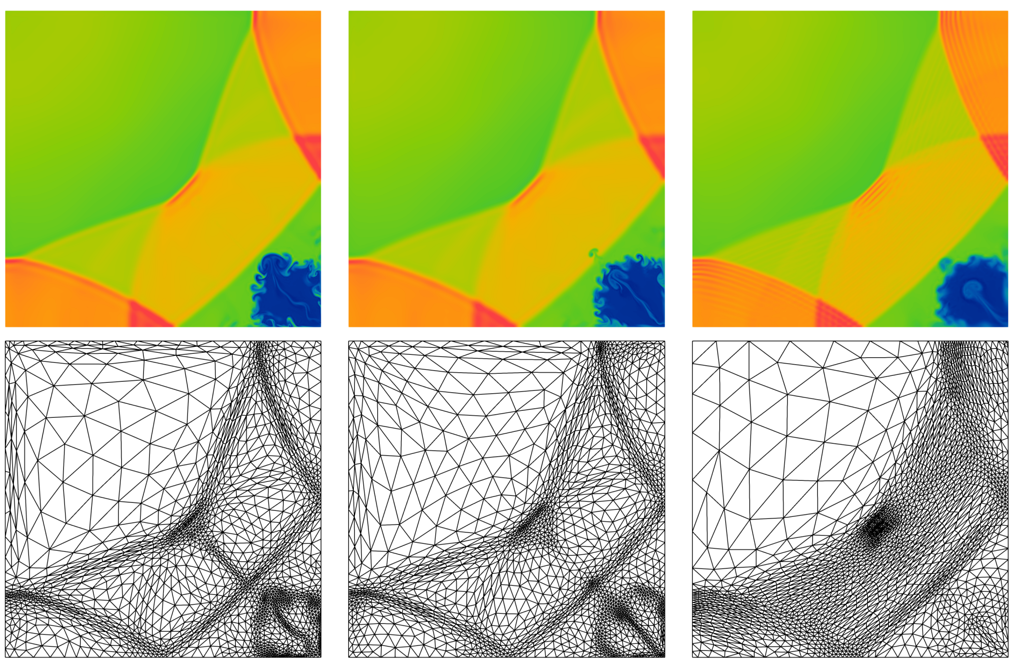
(a) DIRK(3,3) **(b) DIRK(5,4)**

Fig. 12 Implosion problem, $p = 3$, $t = 0.7$, $\Delta t = 0.0025$, Moro's artificial viscosity field.



(a) DIRK(2,2) **(b) DIRK(3,3)** **(c) DIRK(5,4)**

Fig. 13 Implosion problem, contours of density $\rho \in [0.3, 1.2]$ (top) and the corresponding mesh (bottom) at time $t = 0.7$ using $p = 3$ polynomials and $\Delta t = 0.005$ with Hartmann's artificial viscosity



(a) DIRK(3,3), Hartmann AV **(b) DIRK(3,3), Moro AV** **(c) DIRK(5,4), Moro AV**

Fig. 14 Implosion problem, contours of density $\rho \in [0.3, 1.2]$ (top) and the corresponding mesh (bottom) at time $t = 0.7$ using $p = 3$ polynomials and $\Delta t = 0.0025$ with Hartmann's and Moro's artificial viscosity (AV).

E. Shock-Vortex Interaction

To compare the presented method against the classical approach of using a fine static mesh in all time steps, the interaction between a stationary shock and a right moving vortex is considered [28]. The computational domain is taken to be $\Omega = [0, 2] \times [0, 1]$. A stationary shock wave is initialized at $x = 0.5$ with $\mathrm{Ma} = 1.1$ being the Mach number. The left state upstream the shock is assumed to be $(\rho_L, u_L, v_L, P_L) = (1, \mathrm{Ma}\sqrt{\gamma}, 0, 1)$. The right state then corresponds to

$$\rho_R = \rho_L \frac{(\gamma+1)\mathrm{Ma}^2}{2+(\gamma-1)\mathrm{Ma}^2}, \quad u_R = u_L \frac{2+(\gamma-1)\mathrm{Ma}^2}{(\gamma+1)\mathrm{Ma}^2}, \quad v = 0, \quad P_R = P_L\left(1 + \frac{2\gamma}{\gamma+1}(\mathrm{Ma}^2 - 1)\right). \tag{43}$$

The vortex located at $(x_c, y_c) = (0.25, 0.5)$ is defined by perturbation of the left state mean flow in terms of the velocity vector (u, v), temperature $\Theta = P/\rho$, and entropy $S = \ln(P/\rho^\gamma)$. These perturbations are given by

$$(\delta u, \delta v) = \epsilon \exp\left(\alpha(1-\eta^2)\right)\left(\frac{y-y_c}{r_c}, -\frac{x-x_c}{r_c}\right), \tag{44}$$

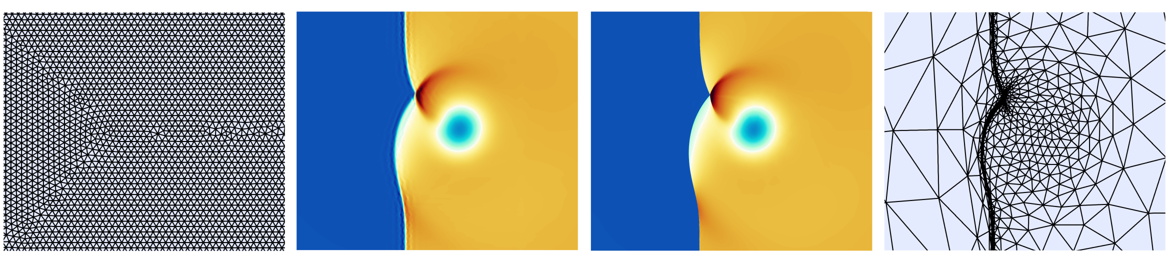
$$\delta\Theta = -\frac{\gamma-1}{4\alpha\gamma}\epsilon^2 \exp\left(2\alpha(1-\eta^2)\right), \tag{45}$$

$$\delta S = 0, \tag{46}$$

where $\eta = r/r_c$ with $r = \sqrt{(x-x_c)^2 + (y-y_c)^2}$, ϵ is the strength of the vortex, α controls the decay rate of the vortex, and r_c is the critical radius for which the vortex has the maximum strength. Based on [29], we set $\epsilon = 0.3$, $\alpha = 0.204$, and $r_c = 0.05$. The reflecting boundary conditions are assumed on the top and bottom of the domain.

The final time of the simulation is set to $T = 1$ and the time step size is taken to be $\Delta t = 0.005$ using the third-order DIRK(3,3) method and the Hartmann's shock-capturing method. We consider two cases. First, the test case has been run on a fixed mesh with 11,000 triangular elements for the whole computation. For the other one, the mesh is initially adapted based on the initial conditions, and then the anisotropic mesh adaptation algorithm is applied in every other time step. The mesh is limited to have 2,000 elements.

The distribution of density for both cases together with the corresponding meshes at time $t = 0.35$ is depicted in Fig. 15. In addition to the improved resolution and suppression of the oscillations near the shock, the computational time for the adapted mesh case including the mesh adaptation procedure is approximately one third of the case with the fixed mesh.



(a) (b) (c) (d)

Fig. 15 Detail of the shock-vortex interaction at time $t = 0.35$. Fixed mesh with 11,000 elements (a) and the corresponding density distribution (b). Contours of density (c) obtained with the adapted mesh (d).

VII. Conclusions and Outlook

Application of anisotropic mesh refinement for time-dependent problems discretized by the HDG method was presented. An algorithm for the conservative interpolation of the numerical solution between two meshes has been incorporated into the overall scheme, and the preliminary results show the spatial and temporal error of the discretization method is not affected.

The numerical tests aiming to compare the properties of three diagonally implicit Runge-Kutta methods coupled with the anisotropic mesh adaptation have shown that the DIRK(3,3) method is superior to the others. Unlike the DIRK(2,2) and DIRK(5,4) methods, it can damp the oscillations present in the numerical solution and help to properly capture the features of the flow.

So far, the presented approach has been investigated also for the Euler equations in test cases with rather weak shock waves where a significant computational cost reduction is achieved. The ongoing work aims to improve the utilized shock-capturing methods, and to develop a bounding algorithm in order to avoid nonphysical numerical solutions, which may occur during the interpolation step, in order to be able to solve problems including strong moving shocks.

Acknowledgments

The work was supported by grant 21–31457S of the Czech Science Foundation and the project SGS–2022–008.

References

[1] Wang, Z. J., Fidkowski, K., Abgrall, R., Bassi, F., Caraeni, D., Cary, A., Deconinck, H., Hartmann, R., Hillewaert, K., Huynh, H. T., Kroll, N., May, G., Persson, P. O., van Leer, B., and Visbal, M., "High-order CFD methods: Current status and perspective," *International Journal for Numerical Methods in Fluids*, Vol. 72, No. 8, 2013, pp. 811–845.

[2] Alauzet, F., and Loseille, A., "A decade of progress on anisotropic mesh adaptation for computational fluid dynamics," *Computer-Aided Design*, Vol. 72, 2016, pp. 13–39.

[3] Grandy, J., "Conservative remapping and region overlays by intersecting arbitrary polyhedra," *Journal of Computational Physics*, Vol. 148, No. 2, 1999, pp. 433–466.

[4] Remacle, J.-F., Li, X., Shephard, M. S., and Flaherty, J. E., "Anisotropic adaptive simulation of transient flows using discontinuous Galerkin methods," *International Journal for Numerical Methods in Engineering*, Vol. 62, No. 7, 2005, pp. 899–923.

[5] Alauzet, F., Loseille, A., and Olivier, G., "Time-accurate multi-scale anisotropic mesh adaptation for unsteady flows in CFD," *Journal of Computational Physics*, Vol. 373, No. C, 2018, p. 28–63.

[6] Alauzet, F., Frey, P., George, P., and Mohammadi, B., "3D transient fixed point mesh adaptation for time-dependent problems: Application to CFD simulations," *Journal of Computational Physics*, Vol. 222, No. 2, 2007, pp. 592–623.

[7] Alauzet, F., "A parallel matrix-free conservative solution interpolation on unstructured tetrahedral meshes," *Computer Methods in Applied Mechanics and Engineering*, Vol. 299, 2016, pp. 116–142.

[8] Rangarajan, A. M., Balan, A., and May, G., "Mesh adaptation and optimization for discontinuous Galerkin methods using a continuous mesh model," *AIAA Modeling and Simulation Technologies Conference*, 2016.

[9] George, P., and Borouchaki, H., *Delaunay triangulation and meshing: Application to finite elements*, Hermès, 1998.

[10] Farrell, P., Piggott, M., Pain, C., Gorman, G., and Wilson, C., "Conservative interpolation between unstructured meshes via supermesh construction," *Computer Methods in Applied Mechanics and Engineering*, Vol. 198, No. 33, 2009, pp. 2632–2642.

[11] Farrell, P. E., and Maddison, J. R., "Conservative interpolation between volume meshes by local Galerkin projection," *Computer Methods in Applied Mechanics and Engineering*, Vol. 200, 2011, pp. 89–100.

[12] Hartmann, R., "Adaptive discontinuous Galerkin methods with shock-capturing for the compressible Navier–Stokes equations," *International Journal for Numerical Methods in Fluids*, Vol. 51, No. 9-10, 2006, pp. 1131–1156.

[13] Moro, D., Nguyen, N. C., and Peraire, J., "Dilation-based shock capturing for high-order methods," *International Journal for Numerical Methods in Fluids*, Vol. 82, No. 7, 2016, pp. 398–416.

[14] Wanner, G., and Hairer, E., *Solving ordinary differential equations II. Stiff and differential-algebraic problems*, Springer-Verlag, Berlin Heidelberg, 1996.

[15] Alexander, R., "Diagonally implicit Runge–Kutta methods for stiff O.D.E.'s," *SIAM Journal on Numerical Analysis*, Vol. 14, No. 6, 1977, pp. 1006–1021.

[16] Cash, J. R., "Diagonally implicit Runge-Kutta formulae with error estimates," *IMA Journal of Applied Mathematics*, Vol. 24, No. 3, 1979, pp. 293–301.

[17] Loseille, A., and Alauzet, F., "Continuous mesh framework part I: Well-posed continuous interpolation error," *SIAM Journal on Numerical Analysis*, Vol. 49, No. 1, 2011, pp. 38–60.

[18] Loseille, A., and Alauzet, F., "Continuous mesh framework part II: Validations and applications," *SIAM Journal on Numerical Analysis*, Vol. 49, No. 1, 2011, pp. 61–86.

[19] Hecht, F., "BAMG: Bidimensional anisotropic mesh generator," *User Guide. INRIA, Rocquencourt*, Vol. 17, 1998.

[20] Dolejší, V., "Anisotropic hp-adaptive method based on interpolation error estimates in the L^q-norm," *Applied Numerical Mathematics*, Vol. 82, 2014, pp. 80–114.

[21] Bartoš, O., Dolejší, V., May, G., Rangarajan, A., and Roskovec, F., "A goal-oriented anisotropic hp-mesh adaptation method for linear convection–diffusion–reaction problems," *Computers & Mathematics with Applications*, Vol. 78, No. 9, 2019, pp. 2973–2993.

[22] Rangarajan, A., May, G., and Dolejší, V., "Adjoint-based anisotropic hp-adaptation for discontinuous Galerkin methods using a continuous mesh model," *Journal of Computational Physics*, Vol. 409, 2020, p. 109321.

[23] Löhner, R., "Robust, vectorized search algorithms for interpolation on unstructured grids," *Journal of Computational Physics*, Vol. 118, No. 2, 1995, pp. 380–387.

[24] Nguyen, N. C., Peraire, J., and Cockburn, B., "An implicit high-order hybridizable discontinuous Galerkin method for linear convection-diffusion equations," *Journal of Computational Physics*, Vol. 228, No. 9, 2009, pp. 3232–3254.

[25] Rudman, M., "Volume-tracking methods for interfacial flow calculations," *International Journal for Numerical Methods in Fluids*, Vol. 24, No. 7, 1997, pp. 671–691.

[26] Liska, R., and Wendroff, B., "Comparison of Several Difference Schemes on 1D and 2D Test Problems for the Euler Equations," *SIAM Journal on Scientific Computing*, Vol. 25, No. 3, 2003, pp. 995–1017.

[27] Barter, G. E., and Darmofal, D. L., "Shock capturing with PDE-based artificial viscosity for DGFEM: Part I. Formulation," *Journal of Computational Physics*, Vol. 229, No. 5, 2010, pp. 1810–1827.

[28] Choi, J. J., "High-order methods for conservation laws employing embedded structured element method on unstructured hexahedral Grid," *22nd AIAA Computational Fluid Dynamics Conference*, 2015.

[29] Zhao, F., Pan, L., and Wang, S., "Weighted essentially non-oscillatory scheme on unstructured quadrilateral and triangular meshes for hyperbolic conservation laws," *Journal of Computational Physics*, Vol. 374, 2018, pp. 605–624.